CARTESIAN MORPHISMS OF EQUIVALENCE RELATIONS GIVE FIBRE PRODUCT SQUARES OF QUOTIENT SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file groupoids.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Lemma lemma-quotient-pre-equivalence (canonical tag 045Y). cited at proof line 4158. Pointwise description of the quotient sheaves; standing throughout.
- Lemma lemma-equivalence-groupoid (canonical tag 0233). cited at proof lines 4171-4173. Identifies the identity section of the groupoid scheme of j'.

Source reference. Original lemma 045Y: sources/stacks-chow/tags.txt, source lines 5453–5453. Cited at proof line 4158 as the standing description of U/R and U'/R' used throughout; load-bearing.

Source reference. Original lemma 0233: sources/stacks-chow/tags.txt, source lines 2728–2728. Cited at proof lines 4171-4173 to identify e'(f(a)) as the identity of the groupoid scheme of j'; load-bearing for the injectivity argument.

Lemma 0.2. *Let S be a scheme. Let $f : (U, R, j) \to (U', R', j')$ be a morphism between equivalence relations over S. Assume that*

$$\begin{array}{ccc} R & \xrightarrow{f} & R' \\ {\scriptstyle s}\downarrow & & \downarrow{\scriptstyle s'} \\ U & \xrightarrow{f} & U' \end{array}$$

is cartesian. For any $\tau \in \{Zariski, étale, fppf, smooth, syntomic\}$ the diagram

$$\begin{array}{ccc} U & \longrightarrow & U/R \\ \downarrow & & \downarrow{\scriptstyle f} \\ U' & \longrightarrow & U'/R' \end{array}$$

is a fibre product square of τ-sheaves.

Date: stacks-project:groupoids.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. By Lemma 045Y the quotient sheaves have a simple description which we will use below without further mention. We first show that

$$U \longrightarrow U' \times_{U'/R'} U/R$$

is injective. Namely, assume $a, b \in U(T)$ map to the same element on the right hand side. Then $f(a) = f(b)$. After replacing T by the members of a τ-covering we may assume that there exists an $r \in R(T)$ such that $a = s(r)$ and $b = t(r)$. Then $r' = f(r)$ is a T-valued point of R' with $s'(r') = t'(r')$. Hence $r' = e'(f(a))$ (where e' is the identity of the groupoid scheme associated to j', see Lemma 0233). Because the first diagram of the lemma is cartesian this implies that r has to equal $e(a)$. Thus $a = b$.

Finally, we show that the displayed arrow is surjective. Let T be a scheme over S and let $(a', \overline{b})$ be a section of the sheaf $U' \times_{U'/R'} U/R$ over T. After replacing T by the members of a τ-covering we may assume that $\overline{b}$ is the class of an element $b \in U(T)$. After replacing T by the members of a τ-covering we may assume that there exists an $r' \in R'(T)$ such that $a' = t(r')$ and $s'(r') = f(b)$. Because the first diagram of the lemma is cartesian we can find $r \in R(T)$ such that $s(r) = b$ and $f(r) = r'$. Then it is clear that $a = t(r) \in U(T)$ is a section which maps to $(a', \overline{b})$. $\square$